

$$\omega_2^{(1)} \Rightarrow 0 - (0.1)(-5) = 0.5$$

$$b^{(1)} \Rightarrow 0 - (0.1)(-5) = 0.5$$

Q1.4.

forward pass epoch 2. $\rightarrow$ $s_1 \rightarrow \hat{y}_1 \Rightarrow (0.8) \times 1 + (0.5) \times 1 + 0.5 \Rightarrow 1.8$

$s_2 \rightarrow \hat{y}_2 \Rightarrow (0.8) \times 2 + (0.5) \times 1 + 0.5 \Rightarrow 2.6$

Using $\omega_1^{(1)} \Rightarrow 0.8$, $\omega_2^{(1)} \Rightarrow 0.5$, $b^{(1)} \Rightarrow 0.5$.

$$e_i \Rightarrow \hat{y}_i - y_i \rightarrow s_1 \rightarrow (1.8 - 2) \Rightarrow -0.2$$

$$s_2 \rightarrow (2.6 - 3) \Rightarrow -0.4$$

$$\rightarrow \frac{\partial L}{\partial \omega_1} \Rightarrow \frac{2}{2} [(-0.2) \times 1 + (-0.4) \times 2] \Rightarrow -1.0$$

$$\rightarrow \frac{\partial L}{\partial \omega_2} \Rightarrow \frac{2}{2} [(-0.2) \times 1 + (-0.4) \times 1] \Rightarrow -0.6$$

$$\rightarrow \frac{\partial L}{\partial b} \Rightarrow \frac{2}{2} [(-0.2) + (-0.4)] \Rightarrow -0.6$$

$$\omega_1^{(2)} \Rightarrow 0.8 - (0.1)(-1) \Rightarrow 0.9$$

$$\omega_2^{(2)} \Rightarrow 0.5 - (0.1)(-0.6) \Rightarrow 0.56$$

$$b^{(2)} \Rightarrow 0.5 - (0.1)(-0.6) \Rightarrow 0.56$$

Forward pass Epoch 2 $\rightarrow$ $s_1 \rightarrow \hat{y}_1 \Rightarrow (0.9) \times 1 + (0.56) \times 1 + 0.56 \Rightarrow 2.02$

$\hat{y}_2 \Rightarrow (0.9) \times 2 + (0.56) \times 1 + 0.56 \Rightarrow 2.92$

Using $\omega_1^{(2)} \Rightarrow 0.9$, $\omega_2^{(2)} \Rightarrow 0.56$, $b^{(2)} \Rightarrow 0.56$.

$e_1 \Rightarrow 2.02 - 2 \Rightarrow 0.02$ $e_2 \Rightarrow 2.92 - 3 \Rightarrow -0.08$

$$\frac{\partial L}{\partial \omega_1} \Rightarrow \frac{2}{2} [0.02 \times 1 + (-0.08) \times 2] \Rightarrow -0.14$$

$$\frac{\partial L}{\partial \omega_2} \Rightarrow \frac{2}{2} [0.02 \times 1 + (-0.08) \times 1] \Rightarrow -0.06$$

$$\frac{\partial L}{\partial b} \Rightarrow \frac{2}{2} [0.02 + (-0.08)] \Rightarrow -0.06$$

$$\omega_1^{(3)} \Rightarrow 0.9 - (0.1)(-0.14) \Rightarrow 0.914$$

$$\omega_2^{(3)} \Rightarrow 0.56 - (0.1)(-0.06) \Rightarrow 0.566$$

$$b^{(3)} \Rightarrow 0.56 - (0.1)(-0.06) \Rightarrow 0.566$$